Sliding Mode State Observer for 2−DOF Twin Rotor MIMO System

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***Abstract*— This paper presents a sliding mode state observer for the 2−DOF twin rotor MIMO (multi-input-multi-output) system which belongs to a class of inherently nonlinear systems. Design parameters are selected such that on the defined switching surface, asymptotically stable sliding mode is always generated. Robust sliding and global asymptotic stability conditions are derived by using Lyapunov method. The unknown nonlinearities are estimated and the state estimation errors tend to zero asymptotically.**

***Index Terms*—Nonlinear systems, sliding mode technique, state observer, twin rotor MIMO system.**

I. Introduction

MANY of the theoretical developments in the area of sliding mode control systems assume that the system state vector is available for use by the control scheme. In order to exploit these control strategies, a suitable estimate of the state vector may be constructed for use in the original control law.

The observer design of an experimental propeller setup called the twin rotor multi-input-multi-output system (TRMS) is proposed. The TRMS [1] is a laboratory setup designed for control experiments. In certain aspects, its behavior resembles that of a helicopter which is typically described as having unstable, nonlinear and coupled dynamics. The modeling and controller design of TRMS has been addressed in the literature [2-5].

Several authors have proposed SMO (sliding-mode observer) design methods [6-9]. The purpose of a state observer is to estimate the unavailable state variables of a plant. The idea of using a dynamical linear system to generate estimates of the plant states can be traced to Luenberger [6], which is well known. The Luenberger observer performs well when the plant dynamics are completely known. The state-feedback-based design of SMC assumes that all the plant states are directly accessible. However, in real systems, all the states are seldom available. One of the solutions is to use an observer.

In the SMO, the error between the observer output and the system output is fed back via a discontinuous switched signal instead of feeding it back linearly. The SMO has a unique feature of generating sliding mode on the error between the measured plant output and the observed output. The effectiveness of the methodology for the observer design for nonlinear systems was considered in [7-8].

Spurgeon describes an overview of linear and nonlinear SMOs in her survey paper [9]. The method of Walcott and Zak [10] requires a symbolic manipulation package to solve the design problem. State observation of nonlinear dynamical systems is a topic of interesting discussion in the literature [11-12]. Thau [11] incorporates the nonlinearities of the plant into the dynamics of his observer design and requires that the nonlinearities be Lipschitz in the states.

The SMO design problem for uncertain dynamical systems subject to external disturbances has been a topic of considerable interest of several authors. There are several observers successfully designed by Utkin [14], Walcott and Zak [15], Walcott, et al. [16], Zak, et al. [17], Edwards and Spurgeon [18,19], Slotine, et al. [20], Watanabe, et al. [21], Hachimoto, et al. [22]; etc., where Lyapunov method has been used to formulate sliding mode observers design which guarantees that the state estimation errors converge to zero asymptotically in the presence of matched uncertainties.

In this paper, a high-gain SMO is proposed to estimate the states using Walcott & Zak observer design approach. A nonlinear state observer for 2-DOF twin rotor MIMO system is presented using sliding mode technique. The present observer does not necessitate exact knowledge of the system nonlinearities. This aim is accomplished by utilizing techniques prevalent in variable structure systems (VSS) theory.

The remainder of the paper is arranged as follows. In Section II, the TRMS system is introduced and the parameters of the system specified. The problem statement is introduced in Section III. In Section IV, the sliding mode observer design is given. The observer performance is demonstrated in Section V by providing simulation results on the TRMS. The simulation results reveal the advantages of the proposed observer and the effect of learning rate. Finally Concluding remarks are made in the section VI.

II. Modeling of 2-DOF TRMS

The TRMS mechanical unit has two rotors placed on a beam together with a counterbalance whose arm with a weight at its end is fixed to the beam at the pivot and it determines a stable equilibrium position as shown in the fig.1. The beam is pivoted on its base in such a way that it can rotate freely both in the horizontal and vertical planes. Either the horizontal or the vertical degree of freedom can be restricted to 1 degree of freedom using nylon screws found near pivot point. At both

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ends of the beam there are rotors (the main and tail rotors) driven by dc motors. The main rotor produces a lifting force allowing the beam to rise vertically making a rotation around the pitch axis. While, the tail rotor is used to make the beam turn left or right around the yaw axis. The whole unit is attached to the tower allowing for safe helicopter control experiments. Apart from the mechanical units, the electrical unit (placed under the tower) plays an important role for TRMS control. It allows for measured signals transfer to the PC and control signal application via an I/O card. The mechanical and electrical units provide a complete control system setup. This device is a multivariable, nonlinear and strongly coupled system, with degrees of freedom on the pitch and yaw angle denoted by ψ and φ respectively.



Fig.1 The twin rotor MIMO system

The state of the beam is described by four process variables: horizontal and vertical angles measured by position sensors fitted at the pivot, and two corresponding angular velocities. Two additional state variables are the momentum of the dc motors. In a normal helicopter, the aerodynamic force is controlled by changing the angle of attack. The laboratory setup in Fig. 1 is so constructed that the angle of attack is fixed. The aerodynamic force is controlled by varying the speed of the rotors. Therefore, the control inputs are the supply voltage of the dc motors. A change in the voltage value results in a change in the rotation speed of the propeller. This further results change in the corresponding position of the beam.

The momentum equation for the vertical movement is given as [1]

$$I_1\ddot{\psi} = M_1 - M_{FG} - M_{B\psi} - M_G, \tag{1}$$

where,

The nonlinear static characteristic

$$M_1 = a_1\tau_1^2 + b_1\tau_1, \tag{2}$$

Gravity momentum

$$M_{FG} = M_g \sin\psi, \tag{3}$$

Friction forces momentum

$$M_{B\psi} = B_{1\psi}\dot{\psi} - \frac{0.0326}{2}\sin 2\psi\dot{\varphi}^2 \tag{4}$$

and Gyroscopic momentum

$$M_G = k_{gy}M_1\dot{\varphi}\cos\psi \tag{5}$$

The motor and the electric control circuit are approximated by a first order transfer function thus in Laplace domain the motor momentum is described by

$$\tau_1 = \frac{k_1}{T_{11}s + T_{10}}u_1 \tag{6}$$

The momentum equation for the vertical movement is given as

$$I_2\ddot{\varphi} = M_2 - M_{B\varphi} - M_R \tag{7}$$

where the nonlinear static characteristic

$$M_2 = a_2\tau_2^2 + b_2\tau_2 \tag{8}$$

Friction forces momentum

$$M_{B\varphi} = B_{1\varphi}\dot{\varphi} \tag{9}$$

and M_R is the cross reaction momentum approximated by,

$$M_R = \frac{k_c\left(T_0 s + 1\right)}{\left(T_p s + 1\right)}M_1 \tag{10}$$

Again the DC motor with the electrical circuit is given by

$$\tau_2 = \frac{k_2}{T_{21}s + T_{20}}u_2 \tag{11}$$

The complete dynamics of the TRMS system (1-11) can be approximately represented in the state-space form as follows:

$$\left.\begin{aligned}
&\frac{d}{dt}\psi = \dot{\psi}\\
&\frac{d}{dt}\dot{\psi} = \frac{a_1}{I_1}\tau_1^2 + \frac{b_1}{I_1}\tau_1 - \frac{M_g}{I_1}\sin\psi + \frac{0.0326}{2I_1}\sin\left(2\psi\right)\dot{\varphi}^2\\
&\qquad - \frac{B_{1\psi}}{I_1}\dot{\psi} - \frac{k_{gy}}{I_1}a_1\cos\left(\psi\right)\dot{\varphi}\tau_1^2 - \frac{k_{gy}}{I_1}b_1\cos\left(\psi\right)\dot{\varphi}\tau_1\\
&\frac{d}{dt}\varphi = \dot{\varphi}\\
&\frac{d}{dt}\dot{\varphi} = \frac{a_2}{I_2}\tau_2^2 + \frac{b_2}{I_2}\tau_2 - \frac{B_{1\varphi}}{I_2}\dot{\varphi} - \frac{1.75}{I_2}k_c a_1\tau_1^2 - \frac{1.75}{I_2}k_c b_1\tau_1\\
&\frac{d}{dt}\tau_1 = -\frac{T_{10}}{T_{11}}\tau_1 + \frac{k_1}{T_{11}}u_1\\
&\frac{d}{dt}\tau_2 = -\frac{T_{20}}{T_{21}}\tau_2 + \frac{k_2}{T_{21}}u_2
\end{aligned}\right\} \tag{12}$$

The output is given by

$$y = \begin{bmatrix}\psi & \varphi\end{bmatrix}^T \tag{13}$$

where,

ψ : Pitch (elevation) angle

φ : Yaw (azimuth) angle

τ_1 : Momentum of main rotor

τ_2 : Momentum of tail rotor

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The system parameters of the TRMS are given in Table I [1].

TABLE I: TRMS SYSTEM PARAMETERS

Parameters	Values
I_1 = Moment of inertia of vertical rotor	6.8×10^{-2} $kg-m^2$
I_2 = Moment of inertia of horizontal rotor	2×10^{-2} $kg-m^2$
a_1 = Static characteristic parameter	0.0135
b_1 = Static characteristic parameter	0.0924
a_2 = Static characteristic parameter	0.02
b_2 = Static characteristic parameter	0.09
M_g = Gravity momentum	0.32 $N-m$
$B_{1\psi}$ = Friction momentum function parameter	6×10^{-3} $N-m-s/rad$
$B_{1\varphi}$ = Friction momentum function parameter	1×10^{-1} $N-m-s/rad$
k_{gy} = Gyroscopic momentum parameter	0.05 s/rad
k_1 = Motor 1 gain	1.1
k_2 = Motor 2 gain	0.8
T_{11} = Motor 1 denominator parameter	1.1
T_{10} = Motor 1 denominator parameter	1
T_{21} = Motor 2 denominator	1
T_{20} = Motor 1 denominator parameter	1
T_p = Cross reaction momentum parameter	2
T_0 = Cross reaction momentum parameter	3.5
k_c = Cross reaction momentum gain	–0.2

The bound for control signal is set to $-2.5\text{V}\ldots+2.5\text{V}$, [1].

III. PROBLEM STATEMENT

The twin rotor MIMO system described by the state space representation:

$$\begin{aligned}\dot{x} &= Ax + f(x,u)\\ y &= Cx\end{aligned} \tag{14}$$

where $x\in\Re^n,\ u\in\Re^m,\ y\in\Re^p,\ A\in\Re^{n\times n},\ C\in\Re^{p\times n}$ and $p\geq m$ in the addition the matrix C is assumed to be of full rank. The function $f(x,u)$ can be construed as the uncertainties or nonlinearities in the plant.

The dynamic state space representation of Twin rotor MIMO system is given in equation (14), which can be represented as follows:

$$A=\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0\\ 0 & -\frac{B_{1\psi}}{I_1} & 0 & 0 & \frac{b_1}{I_1} & 0\\ 0 & 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & -\frac{B_{1\varphi}}{I_2} & 0 & \frac{b_2}{I_2}\\ 0 & 0 & 0 & 0 & -\frac{T_{10}}{T_{11}} & 0\\ 0 & 0 & 0 & 0 & 0 & -\frac{T_{20}}{T_{21}} \end{bmatrix};$$

$$f(x,u)=\begin{bmatrix} 0\\ \left\{\begin{matrix}\frac{a_1}{I_1}\tau_1^2-\frac{M_g}{I_1}\sin\psi+\frac{0.0326}{2I_1}\sin(2\psi)\dot{\varphi}^2\\ -\frac{k_{gy}}{I_1}a_1\cos(\psi)\dot{\varphi}\tau_1^2-\frac{k_{gy}}{I_1}b_1\cos(\psi)\dot{\varphi}\tau_1\end{matrix}\right\}\\ 0\\ \frac{a_2}{I_2}\tau_2^2-\frac{1.75}{I_2}k_c a_1\tau_1^2-\frac{1.75}{I_2}k_c b_1\tau_1\\ \frac{k_1}{T_{11}}u_1\\ \frac{k_2}{T_{21}}u_2 \end{bmatrix};$$

where $x=\begin{bmatrix}\psi & \dot{\psi} & \varphi & \dot{\varphi} & \tau_1 & \tau_2\end{bmatrix}^T$

For existence purposes, we require that $f(x,u)$ be continuous in x.

The objective is to design an observer with inputs y and u whose output $\hat{x}$ will converge to x i.e., $\lim_{t\to\infty}(\hat{x}-x)=0$.

IV. SLIDING MODE OBSERVER DESIGN

A. Preliminary Assumptions:

Consider the following three assumptions pertaining to the system denoted in (14).

A1: The pair (A,C) is detectable which implies that we can find a matrix $L\in\Re^{n\times p}$ such that $\sigma[A_0]\subseteq C^-$, where $A_0=(A-LC)$ and C^- is the open left-half plane.

A2: There exists a symmetric, positive definite matrix $Q\in\Re^{n\times n}$ and function $g(x,u)$

where $g:\Re_+^1\times\Re^n\times\Re^m\to\Re^p$ such that

$$f(x,u)=P^{-1}C^T g(x,u)$$

where $P\in\Re^{n\times n}$ is the unique, positive definite solution to the Lyapunov equation

$$\left(A_0{}^T P+PA_0\right)=-Q$$

A3: There exists a positive scalar valued function, ρ such that

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$$\left\|g(x,u)\right\| \le \rho(x,u)$$

for all $t \in \Re_+^1$ and $x \in \Re_+^n$ and $u \in \Re_+^m$.

Now consider the following nonlinear observer dynamical equation:

$$\begin{aligned} \dot{\hat{x}} &= A\hat{x} - L\left[\hat{y} - y\right] + P^{-1}C^T\nu \\ \hat{y} &= C\hat{x} \end{aligned} \tag{15}$$

where

$$\nu = \begin{cases} -\rho(x,u)\dfrac{Ce}{\|Ce\|} & if \quad Ce \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

Remark:

The observer design incorporates only the bound of the nonlinearities and/or uncertainties $\rho(x,u)$ and does not require exact knowledge concerning the structure of the plant nonlinearities except that they satisfy assumption *A2*.

B. Stability Analysis:

Let the error difference between the observer estimate and the true state be denoted by

$$e = \hat{x} - x$$

Now we state the ensuring theorem.

Theorem 1: Given system (14) and the observer governed by (15), if assumptions *A1 – A3* are valid, then $\lim_{t\to\infty}(\hat{x} - x) = \lim_{t\to\infty} e = 0$.

Proof: The error difference between the output of the observer and the true state obeys the following equations:

$$\begin{aligned} \dot{e} &= \dot{\hat{x}} - \dot{x} \\ &= A\hat{x} - L\left[\hat{y} - y\right] + P^{-1}C^T\nu - Ax - f(x,u) \\ &= (A - LC)e - P^{-1}C^T g(x,u) + P^{-1}C^T\nu \\ \dot{e} &= A_0 e - P^{-1}C^T g(x,u) - P^{-1}C^T\nu \end{aligned} \tag{16}$$

Consider the following positive definite Lyapunov function candidate

$$V = e^T Pe \tag{17}$$

where P is defined in assumption *A2*.

The time derivative of this Lyapunov function candidate is given by

$$\begin{aligned} \dot{V}(e) = e^T\left(A_0{}^T P + PA_0\right)e - 2e^T P\left(P^{-1}C^T g(x,u)\right) \\ + 2e^T P\left(P^{-1}C^T\nu\right) \end{aligned} \tag{18}$$

this simplifies to

$$\dot{V}(e) = -e^T(Q)e - 2e^T C^T g(x,u) - 2e^T C^T \rho(x,u)\frac{Ce}{\|Ce\|} \tag{19}$$

Taking the Euclidean norm of the last term of (19) and noting assumption *A3*, yields

$$\dot{V}(e) \le -e^T(Q)e - 2\left\|g(x,u)\right\|\|Ce\| - 2\rho(x,u)\|Ce\|$$

$$\dot{V}(e) \le -e^T(Q)e + 2\rho(x,u)\|Ce\| - 2\rho(x,u)\|Ce\| \tag{20}$$

$$\dot{V}(e) \le -e^T(Q)e \tag{21}$$

Therefore, the $\lim_{t\to\infty}(\hat{x} - x) = \lim_{t\to\infty} e = 0$

Theorem 1 shows that error difference between the estimate and the true state asymptotically tends to zero. However, it is desirable to know the rate at which the estimate converges since, if the time response of the observer is of the same order or greater than the system's response time, the observer is of little use in an *observer-controller* configuration.

Dividing inequality (21) by (17) yields:

$$-\frac{\dot{V}(e)}{V(e)} \ge \frac{e^T(Q)e}{e^T(P)e} \tag{22}$$

or

$$V\left(e(t)\right) \le V\left(e_0,(t_0)\right)e^{-\eta(t-t_0)} \tag{23}$$

where η is the minimum eigen value of $\left(P^{-1}Q\right)$. Thus, if we consider $e^T(P)e$ to be a measure of the magnitude of the error, then the error will approach zero in magnitude exponentially, with a rate of decay that is at least as fast $e^{-\eta t}$.

V. Simulation Results

A detailed simulation study of the proposed observer is carried out. Simulation results of the proposed observer show reliable performance and acceptable computation time for real-time implementation.

The designed sliding mode state observer (15) has been implemented on the TRMS system using the following parameters.

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -0.0882 & 0 & 0 & 1.3588 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -5 & 0 & 4.5 \\ 0 & 0 & 0 & 0 & -0.9091 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix};$$

and

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix};$$

Now readily confirm that the pair (A,C) is observable thus, we may arbitrarily assign the spectrum of $A_0 = (A - LC)$. If we select the spectrum $[-99.003,\ -0.9971+1.1737\,i,\ -0.9971-1.1737\,i,\ -98.9849,\ -4.8096,\ -2.2055]$, the co-responding gain matrix L which will satisfy this requirement is

$$L = \begin{bmatrix} 100 & 100 & 0 & 0 & 100 & 0 \\ 0 & 0 & 100 & 100 & 0 & 100 \end{bmatrix}^T;$$

and corresponding A_0 matrix is,

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$$A_0 = \begin{bmatrix} -100 & 1 & 0 & 0 & 0 & 0 \\ -100 & -0.0882 & 0 & 0 & 1.3588 & 0 \\ 0 & 0 & -100 & 1 & 0 & 0 \\ 0 & 0 & -100 & -5 & 0 & 4.5 \\ -100 & 0 & 0 & 0 & -0.9091 & 0 \\ 0 & 0 & -100 & 0 & 0 & -1 \end{bmatrix};$$

Next, we must find a matrix Q to satisfy assumption $A2$. One may easily verify that

$$Q = \begin{bmatrix} 1000 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1000 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1000 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1000 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1000 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1000 \end{bmatrix};$$

and the corresponding positive definite matrix P, which satisfies the Lyapunov equation in assumption $A2$ is

$$P = \begin{bmatrix} 1061.8 & -460.1 & 0 & 0 & -596.7 & 0 \\ -460.1 & 452.7 & 0 & 0 & 18.4 & 0 \\ 0 & 0 & 519.2 & -68.2 & 0 & -446 \\ 0 & 0 & -68.2 & 86.4 & 0 & -9.6 \\ -596.7 & 18.4 & 0 & 0 & 577.5 & 0 \\ 0 & 0 & -446 & -9.6 & 0 & 457 \end{bmatrix};$$

In the observer (15) a small value of $P^{-1}C^T\nu$ is desirable to minimize the chattering effect. A higher value of Q will result in lower value of P^{-1} as is clear from assumption $A2$.

With the completion of the observer design algorithm, we may now prescribe the dynamics of the final observer via (14).

$$\begin{bmatrix} \dot{\hat{x}}_1 \\ \dot{\hat{x}}_2 \\ \dot{\hat{x}}_3 \\ \dot{\hat{x}}_4 \\ \dot{\hat{x}}_5 \\ \dot{\hat{x}}_6 \end{bmatrix} = \begin{bmatrix} -100 & 1 & 0 & 0 & 0 & 0 \\ -100 & -0.0882 & 0 & 0 & 1.3588 & 0 \\ 0 & 0 & -100 & 1 & 0 & 0 \\ 0 & 0 & -100 & -5 & 0 & 4.5 \\ -100 & 0 & 0 & 0 & -0.9091 & 0 \\ 0 & 0 & -100 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \\ \hat{x}_3 \\ \hat{x}_4 \\ \hat{x}_5 \\ \hat{x}_6 \end{bmatrix} + \begin{bmatrix} 100 & 0 \\ 100 & 0 \\ 0 & 100 \\ 0 & 100 \\ 100 & 0 \\ 0 & 100 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} + \begin{bmatrix} 0.067 & 0 \\ 0.0653 & 0 \\ 0 & 0.0708 \\ 0 & 0.0637 \\ 0.0671 & 0 \\ 0 & 0.0704 \end{bmatrix} \begin{bmatrix} -\rho_1(x,u)\,\mathrm{sgn}(\hat{x}_1 - x_1) \\ -\rho_2(x,u)\,\mathrm{sgn}(\hat{x}_3 - x_3) \end{bmatrix}$$

Since, ρ depends upon the physical properties of the system, hence accordingly it is choosen as $\rho_1 = \rho_2 = 2$. The system is operated in the open loop with the main rotor and tail rotor inputs as $u_1 = u_2 = 2.5\sin(0.5t)$. The initial conditions of the plant and observer are $\begin{bmatrix} 0.5 & 0 & 0.5 & 0 & 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ respectively. The inputs and outputs of the TRMS system are given as the input to the observer.

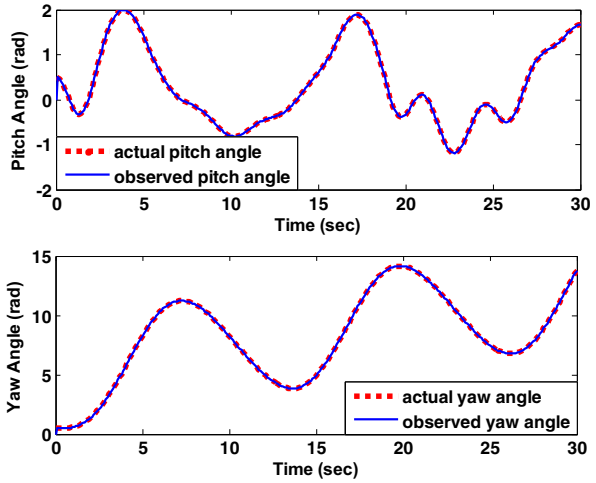


Fig. 2. Actual and estimated states of TRMS system with SMO

The response of the actual state and the observer state is shown in Fig. 2. The efficiency of this observer will depend on the accuracy of the model. Hence, we need a priori knowledge about the system dynamics.

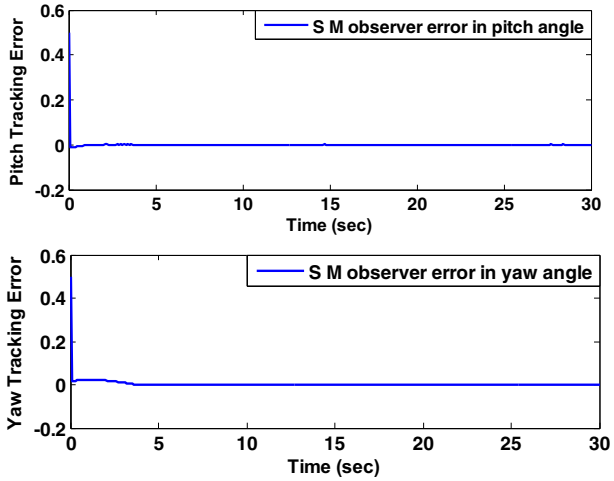


Fig. 3. State estimation error of TRMS system with SMO

The observer error of TRMS system with sliding mode observer is shown in Fig. 3. The error between the actual and the observer states during the steady state is bounded within a small region. The effect of initial conditions is very small in proposed approach and the observer errors converge to zero quickly.

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VI. Concluding Remarks

A nonlinear state observer for 2-DOF twin rotor MIMO system using sliding mode methodology for systems containing completely observable linear parts and bounded nonlinearities or uncertainties is presented in this paper. A minimum estimate for the rate of convergence of the observer error to zero is also given. To test the applicability of the proposed observer in real time is the proposed future scope of work. For real time implementation, the experiments have to be carried out on the real-time 2-DOF TRMS system using MATLAB real-time tool box and Advantech PCI1711 card.

VII. Acknowledgement

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